

School of Computer and Systems Sciences

Jawaharlal Nehru University, New Delhi

Subject: Artificial Intelligence; CS-411

May 2024; Time: 3 hr.

Total Marks: 50

Note: Attempt any five questions

Q1. Answer the following question

(1*10)

- a) Translate 'A car without wheels is not valuable' into predicate logic.
- b) Describe the characteristics of A* and AO* algorithm.
- c) Enumerate the diverse uses of AI.
- d) Provide a concise overview of syntactic processing.
- e) Define rule-based learning.
- f) Define Heuristics.
- g) Explain the concept of an inference engine.
- h) Describe the connection between Intelligence and Knowledge.
- i) Explain the meanings of 'Fact' and 'Rule'.
- j) Outline the criteria used to evaluate search strategies.

Q2. Answer the following questions:

(4*2.5)

- a) Explain Alpha-Beta Pruning with example.
- b) Discuss the role of reasoning in AI. How predicate logic is used in AI to represent Q
- c) What are the differences and similarities between problem-solving and planning?
- d) Explain the state space approach for solving any AI problem

Q3. Answer the following questions

(5*2)

- a) Explain the concept of AI agent and its significance in AI research? List some common applications of Artificial Intelligence in various industries?
- b) Explain A* searching technique in detail with an example. Discuss conditions for the optimality of this technique.

Q4. Answer the following question

(10)

- a) Write about the analysis of search methods.
- b) Define components of AI program.
- c) Mention the categorization of intelligent systems.
- d) Characteristics of a problem.
- e) What are the foundations of AI?

Q5. Consider a state space where the start state is number 1 and each state k has two successors: numbers $2k$ and $2k + 1$.

- a) Draw the portion of the state space for states 1 to 15.
- b) Suppose the goal state is 11. List the order in which nodes will be visited for breadth first search, depth-limited search with limit 3, and iterative deepening search.
- c) How well would bidirectional search work on this problem? What is the branching factor in each direction of the bidirectional search?
- d) Does the answer to (c) suggest a reformulation of the problem that would allow you to solve the problem of getting from state 1 to a given goal state with almost no search?
- e) Call the action going from k to $2k$ Left, and the action going to $2k + 1$ Right. Can you find an algorithm that outputs the solution to this problem without any search at all?

(10)